

JAN SEVA (■■■ ■■■■■)

Comprehensive Project Architecture, Technical Analysis & System Specification

Target Municipality: Gwalior Municipal Corporation

Stack: React 19 + TypeScript + Vite

Version: 1.0.0 Production Ready

Standards: Open311 / DPDP Act 2023

1. Project Overview

Jan Seva (■■■ ■■■■■) is an enterprise-grade civic grievance redressal and municipal operations platform. Unlike legacy portals that act as passive complaint dumpyards, Jan Seva operates as an active, closed-loop civic operating system designed to eliminate ticket duplication, eradicate ghost/fake resolutions, and ensure complete transparency.

The Core Problems Solved:

- **Ticket Duplication & Clutter:** Tens of residents reporting the same pothole or broken streetlight traditionally created dozens of isolated tickets, overloading department queues.
- **Fraudulent & Ghost Resolutions:** Field workers frequently close grievances using old photos, screenshots from messaging apps, or photos taken from depots.
- **DPDP Act Non-Compliance:** Citizen phone numbers and Aadhaar numbers exposed in plain text to field contractors, creating privacy risks.
- **Lack of SLA Accountability:** Grievances disappear into administrative voids without real-time citizen tracking or escalation triggers.

2. Complete End-to-End Workflow

The system enforces a 5-stage citizen reporting wizard linked directly to department dispatch and commissioner command oversight:

Step	Phase	Technical Execution
01	Photo Capture	Camera shutter / gallery picker. Canvas 2D scales to 1600px, matting transparency to white and re-compressing JPEG < 900 KB.
02	Description & Voice	Real-time speech recognition (Web Speech API) with bilingual Hindi/English NLP keyword classification.
03	Identity & Masking	Aadhaar / Mobile OTP challenge. Raw identifiers hashed into HMAC tokens; display masked as XXXX-XXXX-4210.
04	Dual-Location GPS	Device GPS fix combined with OpenStreetMap Nominatim reverse geocoding and fallback landmark search.
05	Screening & Review	Pre-submission fail-open risk pipeline checks dHash image novelty and 150m spatial duplicate clusters.
06	Department Dispatch	Auto-routed to PWD/Sanitation/Water Works/Electrical with computed SLA countdowns and Work Cards.
07	Proof Resolution	Field crew captures on-site fix photo. Proof Engine validates live shutter, GPS < 120m, and dHash originality.

08	Citizen Closure	Resolution photo and verified badge published to citizen tracking timeline with 72h re-open window.
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3. Deep Features Analysis

Feature Name	Internal Implementation	Civic / Admin Benefit
Perceptual Proof Engine	Calculates 64-bit dHash on canvas; validates camera GPS <= 120m from issue.	Eradicates ghost resolutions and duplicate contractor billing.
Spatial Duplicate Grouping	Haversine distance (geoService.ts) clusters reports <= 150m into single root issue.	Prevents ticket duplication; pools citizen support into one high-priority ticket.
Fail-Open Screening	Multi-signal risk score (image reuse, submission bursts) with fail-open guarantee.	Spam and non-civic memes are filtered without blocking real emergencies.
DPDP Privacy Shield	Irreversible HMAC identity tokens (identityService.ts) and client-side masking.	Protects citizen phone numbers and Aadhaar IDs from field contractors.
Bilingual i18n & Voice	Context-aware i18n dictionary (strings.ts) and Web Speech API dictation.	Provides seamless accessibility for non-literate and regional Hindi speakers.
Executive Command Centre	Geospatial heatmaps, live SLA burn-down rates, department escalations, open data.	Gives Municipal Commissioners total operational visibility and control.

4. System Architecture & Technical Stack

Jan Seva is architected as a high-performance, modular React 19 Single Page Application (SPA) leveraging a local-first service layer with asynchronous external integrations:

Layer	Technology	Role in Jan Seva
Frontend UI	React 19.2.8 + React Router 7	Concurrent rendering, route-level code splitting, lazy loading.
Language	TypeScript ~6.0.2	Strict domain type definitions across complaints, proofs, and screening.
Styling System	Vanilla CSS Custom Properties	Zero-runtime overhead design system with dark mode and micro-interactions.
Bundling	Vite 8.2.2 (ESBuild / Rollup)	Sub-second HMR and tree-shaken production asset packaging.
Analytics	@vercel/analytics 2.0.1	Real-time client telemetry, Web Vitals, and route conversion tracking.
Mapping & Geo	OpenStreetMap Nominatim API	Live reverse geocoding, address structuring, and landmark lookup.
Verification	HTML5 Canvas 2D + dHash	Client-side image scaling, thumbnailing, and perceptual difference hashing.
Storage Engine	LocalStorage / IndexedDB Engine	Versioned storage schemas with migrations, seed hydration, and optimistic updates.

5. Module-wise Breakdown

- 1. Public Citizen Portal (`/``, `/report``, `/track``):** Frictionless 5-step reporting wizard with draft persistence, instant verification, live timeline tracking, and feedback submission.

- 2. **Department Operations Portal** (`/dept/*`): Role-based dispatch workspaces for PWD, Sanitation, Water Works, Electrical, and Urban Infra. Includes Work Cards, Map View, and Shutter Proof validation.
- 3. **Municipal Admin Command Centre** (`/admin/*`): Executive dashboard for Municipal Commissioners featuring real-time city health index, SLA burn-down rates, cross-department transfers, and Open311 / GeoJSON export.
- 4. **Moderation & Audit Engine** (`/admin/moderation``, `/admin/ledger``): Append-only chronological audit log tracking every state change, reason, and actor ID with moderation review queues.

6. Data Architecture, Security & DPDP Compliance

Data Collections: ``jan_seva_complaints_v1``, ``jan_seva_issues_v1``, ``jan_seva_audit_log_v1``, ``jan_seva_moderation_cases_v1``, ``jan_seva_evidence_hashes_v1``, ``jan_seva_portal_session_v1``.
Security Controls: Role-Based Access Control (RBAC), exponential login throttling, masked citizen records, DPDP Act 2023 compliance, tamper-evident audit logs, and fail-open screening pipelines.

7. Viva & Project Review Presentation Summary

Executive Presentation Script:
"Jan Seva transforms civic grievance redressal from a broken ticketing form into a verifiable, closed-loop civic operating system. Our three core technical pillars are: (1) Dual-Location & Spatial Clustering (150m) to eliminate ticket duplication, (2) Perceptual Proof Verification (dHash + GPS <= 120m) to stop ghost/fake closures, and (3) DPDP Act 2023 Compliant Identity Masking. Built with React 19, TypeScript, and Vite, the platform features offline resilience, English/Hindi localization, Open311 compliance, and over 100 automated self-test verification suites."